



**iRA
Synergy**
— Private Limited —
Nashik, Maharashtra, India

iRA's RENEW
CLEANER • GREENER • SMARTER



SUSTAINABLE SOLUTIONS
FOR A BETTER TOMORROW

SMART SOLAR BENCH

NEXT-GENERATION SMART CITY FURNITURE

Powered entirely by solar energy, the Solar Smart Bench provides device charging, WIFI connectivity, ambient lighting, and environmental monitoring for modern public spaces.



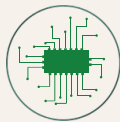
**SOLAR
POWERED**



**USB
CHARGING**



**PUBLIC
WIFI**



**SMART
SENSORS**



**SUSTAINABLE
DESIGN**



**RENEWABLE
ENERGY POWERED**



**SMART
CONNECTIVITY**



**LOW
MAINTENANCE**



**DURABLE
CONSTRUCTION**

SMART SOLAR BENCH

BACK VIEW



FRONT VIEW



KEY FEATURES



**WIRELESS
CHARGING**



**WEATHER
RESISTANCE**



**SMART
LIGHTINGS**



**COMFORTABLE
ERGONOMIC
SEATING**



**SMART
MONITORING**

TECHNICAL SPECIFICATIONS

ABOUT THE PRODUCT

The Solar Smart Bench is a next-generation urban seating solution that brings connectivity and renewable energy to public spaces. Fully powered by integrated solar panels, it provides free Wi-Fi, wireless device charging, USB ports, and atmospheric sensors. Designed with durable, weather-resistant materials, it's perfect for parks, university campus, smart city squares, and corporate campuses.

Additionally, the Solar Smart Bench enhances the overall user experience by creating a comfortable and technology-enabled public environment. By combining sustainable energy generation with smart city infrastructure, it promotes digital accessibility, encourages outdoor engagement, and supports modern urban lifestyles while reducing dependence on conventional power sources.



TECHNICAL SPECIFICATIONS

PARAMETER	DETAILS
 Model	Solar Smart Bench
 Power Source	Monocrystalline Solar Panel
 Solar Panel Capacity	150W – 450W
 Battery Type	LiFePO4 / Lithium-Ion
 Battery Capacity	20Ah – 100Ah
 Charging Ports	USB-A, USB-C

PARAMETER	DETAILS
 Wireless Charging	Qi Standard Wireless Charger
 Wi-Fi Connectivity	4G/5G Router Enabled
 Internet Range	Up to 50 Meters
 LED Lighting	Integrated LED Strip / Downlights
 Sensor Options	Air Quality, Temperature, Humidity, Noise, PM2.5
 Display System	Optional LCD / Digital Display



Energy Independent Operation



Carbon Emission Reduction



Smart City Integration



Public Utility Enhancement



Future-Ready Infrastructure

APPLICATION AREAS



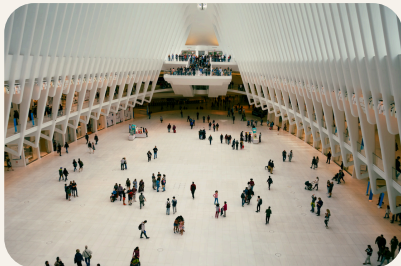
PUBLIC PARKS & RECREATIONAL SPACES



EDUCATIONAL INSTITUTIONS & CAMPUSES



CORPORATE & IT CAMPUSES



TRANSPORTATION HUBS



SMART CITIES & URBAN PLAZAS



TOURIST DESTINATIONS & WATERFRONTS



BUILD TO MEET GOVERNMENT STANDARDS



Complies with CPCB Guidelines for Animal Waste Incineration



Aligned with Solid Waste Management Rules, 2016



Supports Clean India & Swachh Bharat Abhiyan



Eligible for State & Central Government Approvals & Subsidies*

*As per applicable policies.



SAFE & HYGIENIC

Complies with CPCB Guidelines for Animal Waste Incineration



HIGH EFFICIENCY

Advanced 5-chamber technology for maximum efficiency



COST EFFECTIVE

Low operating cost with high performance



COMPLIANT SOLUTION

Meets CPCB norms & environmental standards

SAFETY FEATURES



Electrical Overload Protection

Protects charging ports, batteries, and electronic components from voltage fluctuations, overloads, and short circuits.



Advanced Battery Management System (BMS)

Continuously monitors battery health, temperature, charging cycles, and power distribution for safe operation.



IP65 Weatherproof Electronics Enclosure

Sealed electrical compartments protect internal components from rain, dust, moisture, and outdoor environmental conditions.



Fire-Resistant Electrical Components

Utilizes flame-retardant wiring, insulated connectors, and certified electrical systems to minimize fire risks.



Anti-Vandal & Tamper-Resistant Design

Robust steel construction with secure access panels protects equipment from unauthorized access and vandalism.



ADDITIONAL SAFETY HIGHLIGHTS



SHORT-CIRCUIT
PROTECTION



REMOTE SYSTEM
HEALTH MONITORING



ANTI-VANDAL
CONSTRUCTION



SAFE LOW-VOLTAGE
OPERATION

CERTIFICATES & COMPLIANCE

Manufactured in accordance with national and international standards to ensure quality, safety, and environmental responsibility.



SUSTAINABLE SOLUTIONS FOR A CLEANER TOMORROW

CLEANER ● GREENER ● SMARTER



ABOUT US

iRA Synergy Private Limited is committed to delivering innovative and sustainable waste management solutions that protect health, preserve the environment, and build a cleaner tomorrow.



OUR APPROACH



SUSTAINABLE BY DESIGN

Environmentally responsible solutions designed for long-term impact.



SAFE & RELIABLE

Engineered for safety, reliability and consistent performance.



ADVANCED TECHNOLOGY

Innovative technology for efficient and effective waste management.



COST EFFECTIVE

Low operating costs and minimal maintenance for long-term savings.



OUR COMMITMENT

We aim to create a healthier environment through responsible waste management, continuous innovation and a strong commitment to quality, safety and sustainability.



Clean Environment



Smart Solutions



Green Future

GET IN TOUCH



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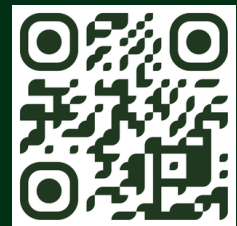


Nashik, Maharashtra, India

FOLLOW US



Let's work together for
a cleaner, greener future.



SCAN TO SEE
our solutions in action